

USTECH
INNOVATIONS

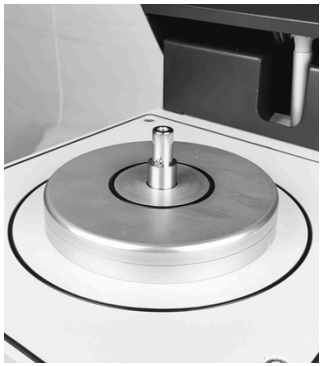
USTECH Tornado+
Centrifugal Rotor Mill for Tough & Temperature-Sensitive Samples

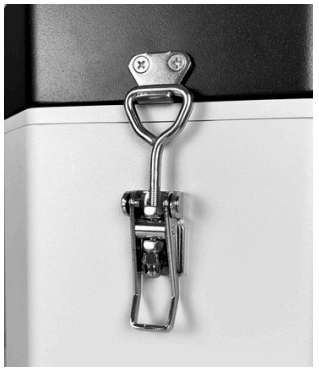


USTECH Tornado+

The Coolest Answer to Tough Samples

The USTECH Tornado+ uses innovative water-cooling technology to effortlessly process high-fat, high-moisture, and high-fiber samples. It ensures rapid, homogeneous results for grains, seeds, and feed while protecting volatile components in temperature-sensitive materials.







Advanced Water-Cooling

Loss-free grinding for temperature-sensitive samples.

Heat generated during grinding degrades high-moisture and high-fat samples, losing volatile components. The USTECH Tornado+ features an integrated water-cooling system that eliminates thermal damage by keeping samples and tools constantly cool.

Maximum Operational Safety

A risk-free working environment

The unique and robust housing architecture ensures maximum operational safety while delivering effortless handling. Thanks to the easy-to-install ring sieve and secure locking mechanism, the system becomes ready for the next analysis in seconds.

Easy Installation & Fast Cleaning

Effortless maintenance with its easy-to-install ring sieve

The USTECH Tornado+ features a practical design for easy sample discharge from the grinding chamber. The easily removable rotor and ring sieve allow fast, thorough cleaning between analyses, minimizing cross-contamination risks and maximizing operational efficiency.



Technical Specifications

Sample Type	Grain, cereal, seed, feed (up to 6 mm pellets)
Sample Size	50-100 g
Grinding Speed	9000 rpm
Particle Size	Depending on the sieve type
Safety	Microswitch
Cleanin	Manual
Water Supply	2L/min, 10-15 °C
Dimentions	255x370x500 mm
Power Supply	220-240 V, 50-60 Hz

Mastering Complex Sample Matrices

Redefining sample preparation for your laboratory's toughest demands.



Grains & Cereals

Achieving complete homogeneity in complex cereal matrices is essential for accurate downstream analysis. The USTECH Tornado+ effortlessly processes wheat, barley, and mixed grains, providing a perfectly uniform sample that guarantees standardized results every cycle.



High-Fat & Oilseeds

Grinding high-fat samples often generates heat that degrades volatile components and alters nutritional profiles. Powered by a continuous water-cooling system that maintains optimal low temperatures, the Tornado+ eliminates thermal damage to fully preserve the biochemical integrity of sensitive seeds.



High-Moisture & Fibrous

Working with high-moisture and fibrous materials can cause blockages in standard mills. With its easily interchangeable ring sieves, you can select the exact particle size required, ensuring smooth, continuous, and loss-free grinding for the most challenging samples.



Animal Feed & Pellets

Formulation consistency in feed requires highly reproducible sample preparation. Engineered specifically to handle complex feed mixtures, it effortlessly processes tough pellets up to 6 mm in diameter, allowing you to maintain strict quality control without workflow interruptions.

Our Core Commitments to You

"The Standard in Spectroscopy" is our commitment to your success. We provide cutting-edge analytical solutions so you can focus entirely on optimizing production, streamlining workflows, and ensuring uncompromised product quality.



Innovative

We leverage advanced technological expertise to develop state-of-the-art solutions—like direct measurement systems—that redefine laboratory standards.



Reliable

We guarantee precision and reproducibility, ensuring consistent, lab-grade results you can trust in every analysis.



Effortless

We simplify your daily workflow with intuitively designed instruments requiring zero-to-minimal sample preparation.



Cost-Effective

Maximize economic benefit by completely eliminating the need for expensive consumables, toxic reagents, and time-consuming methods.



Safe & Clean

We prioritize operator safety and the environment through robust housing architectures and chemical-free analytical processes.



Fast

Empowering rapid decisions with instant insights. We deliver accurate results in under a minute to keep your production moving.

CONTACT US

USTECH INNOVATIONS

600 Park Offices Dr, Ste 300
Durham, NC 27713, USA
info@ustechinnovations.com

www.ustechinnovations.com